

Assignment 2: Power BI – DAX and Data Visualization

(Data Analytics)

1

Assignment Tasks:

Dataset Description :

There are three datasets in the Given Data set:

- List of orders:

*Attributes: Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit

- Order Details

*Attributes: Order ID, Category, Sub-Category, Quantity, Amount, Profit

- Sales Target

*Attributes: Category, Month of Order Date, Target

Tasks :

DAX Calculations:

1. Calculated Columns:

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Auto recovery contains some recovered files that haven't been opened. View recovered files

Category Type = CONCATENATE('Order Details'[Category] , 'Order Details'[Sub-Category])

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25602	561	212	3	Clothing	Saree	ClothingSaree
B-25602	119	-5	8	Clothing	Saree	ClothingSaree
B-25603	193	-166	3	Clothing	Saree	ClothingSaree
B-25604	157	5	9	Clothing	Saree	ClothingSaree
B-25605	75	0	7	Clothing	Saree	ClothingSaree
B-25609	25	-5	4	Clothing	Saree	ClothingSaree
B-25610	43	0	3	Clothing	Saree	ClothingSaree
B-25611	160	-59	2	Clothing	Saree	ClothingSaree
B-25613	1603	0	9	Clothing	Saree	ClothingSaree
B-25619	353	90	8	Clothing	Saree	ClothingSaree
B-25622	534	0	3	Clothing	Saree	ClothingSaree
B-25623	149	-87	4	Clothing	Saree	ClothingSaree
B-25625	635	-349	5	Clothing	Saree	ClothingSaree
B-25628	24	-9	4	Clothing	Saree	ClothingSaree
B-25633	711	-8	4	Clothing	Saree	ClothingSaree
B-25635	382	30	3	Clothing	Saree	ClothingSaree
B-25636	637	113	5	Clothing	Saree	ClothingSaree
B-25640	122	-47	4	Clothing	Saree	ClothingSaree
B-25646	20	-8	2	Clothing	Saree	ClothingSaree
B-25647	42	-6	4	Clothing	Saree	ClothingSaree
B-25648	55	-26	4	Clothing	Saree	ClothingSaree

Order Details (1,500 rows) Column: Category Type (17 distinct values)

Update available (click to down)

Data

Search

- List of Orders
 - City
 - CustomerName
- Order Date
 - Order ID
 - State
- Order Details
 - Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Sub-Category
- Sales Target

The Power BI Data Analytics is performed using the DAX Functions and Data Visualization . From the given Dataset the Dataset csv Files are uploaded and Renamed.

Dax Calculation is Performed with Calculated Column,Calculated Table And Calculated Measure.

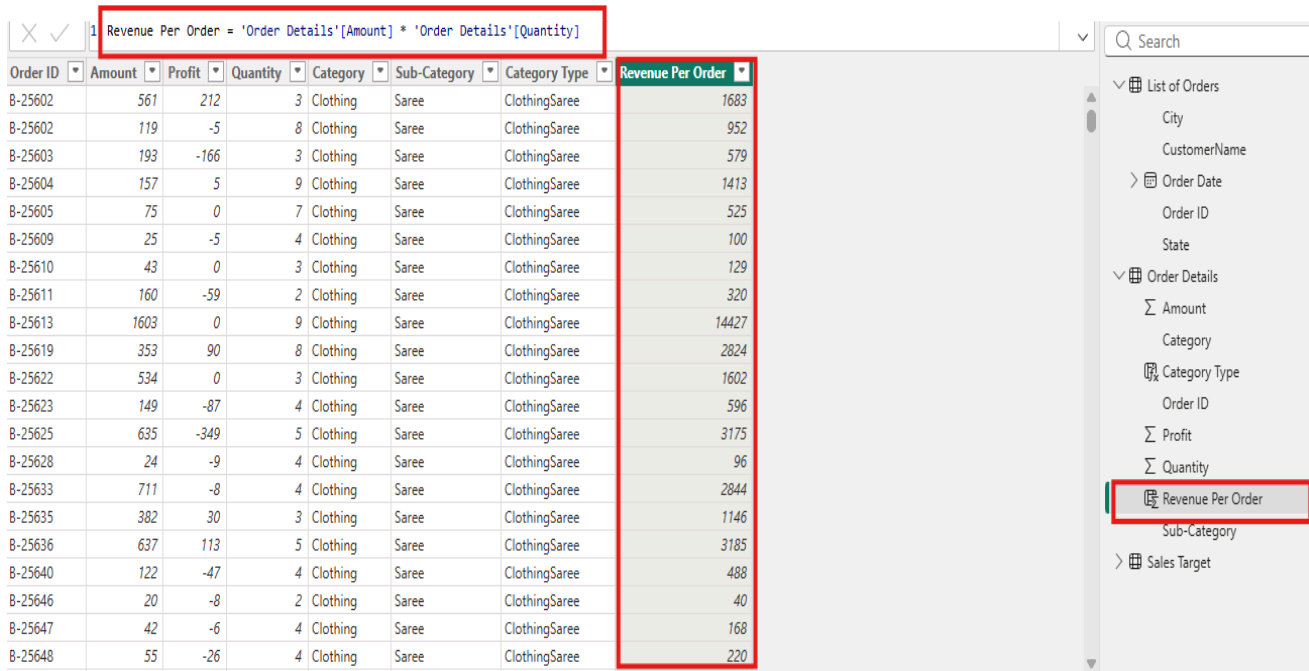
In this Above Image Representing the Calculated Column to create a new column Named as Category Type.

To create a new Calculated column go to Table Tools→select Calculated Column.

After that, write the formula as Category Type = `CONCATENATE('Order Details'[Category], 'Order Details'[Sub-Category])` which is created using the concatenate function.

Then the Category column is created.

● **Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.**



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order
B-25602	561	212	3	Clothing	Saree	ClothingSaree	1683
B-25602	119	-5	8	Clothing	Saree	ClothingSaree	952
B-25603	193	-166	3	Clothing	Saree	ClothingSaree	579
B-25604	157	5	9	Clothing	Saree	ClothingSaree	1413
B-25605	75	0	7	Clothing	Saree	ClothingSaree	525
B-25609	25	-5	4	Clothing	Saree	ClothingSaree	100
B-25610	43	0	3	Clothing	Saree	ClothingSaree	129
B-25611	160	-59	2	Clothing	Saree	ClothingSaree	320
B-25613	1603	0	9	Clothing	Saree	ClothingSaree	14427
B-25619	353	90	8	Clothing	Saree	ClothingSaree	2824
B-25622	534	0	3	Clothing	Saree	ClothingSaree	1602
B-25623	149	-87	4	Clothing	Saree	ClothingSaree	596
B-25625	635	-349	5	Clothing	Saree	ClothingSaree	3175
B-25628	24	-9	4	Clothing	Saree	ClothingSaree	96
B-25633	711	-8	4	Clothing	Saree	ClothingSaree	2844
B-25635	382	30	3	Clothing	Saree	ClothingSaree	1146
B-25636	637	113	5	Clothing	Saree	ClothingSaree	3185
B-25640	122	-47	4	Clothing	Saree	ClothingSaree	488
B-25646	20	-8	2	Clothing	Saree	ClothingSaree	40
B-25647	42	-6	4	Clothing	Saree	ClothingSaree	168
B-25648	55	-26	4	Clothing	Saree	ClothingSaree	220

A new Calculated column is created using the Ribbon Table tools→select the Calculated column →Click in that Calculated column then there will be a new column created and then write the formula as Revenue Per Order=`'Order Details'[Amount] * 'Order Details'[Quantity]`

- **Create a Calculated Column to Categorize Sales:** Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

1 Sales Category = If('Order Details'[Amount] > 288,"Above Average","Below Average")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	ClothingSaree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	ClothingSaree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	ClothingSaree	579	Below Average
B-25604	157	5	9	Clothing	Saree	ClothingSaree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	ClothingSaree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	ClothingSaree	100	Below Average
B-25610	43	0	3	Clothing	Saree	ClothingSaree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	ClothingSaree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	ClothingSaree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	ClothingSaree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	ClothingSaree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	ClothingSaree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	ClothingSaree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	ClothingSaree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	ClothingSaree	2844	Above Average
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B-25640	122	-47	4	Clothing	Saree	ClothingSaree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	ClothingSaree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	ClothingSaree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	ClothingSaree	220	Below Average

Order Details (1,500 rows) Columns: Sales Category (2 distinct values)

Search

- List of Orders
 - City
 - CustomerName
- Order Date
 - Order ID
 - State
- Order Details
 - Amount
 - Average Amount
 - Category
 - Category Type
 - Order ID
 - Profit
 - Quantity
 - Revenue Per Order
 - Sales Category**
 - Sub-Category
- Sales Target

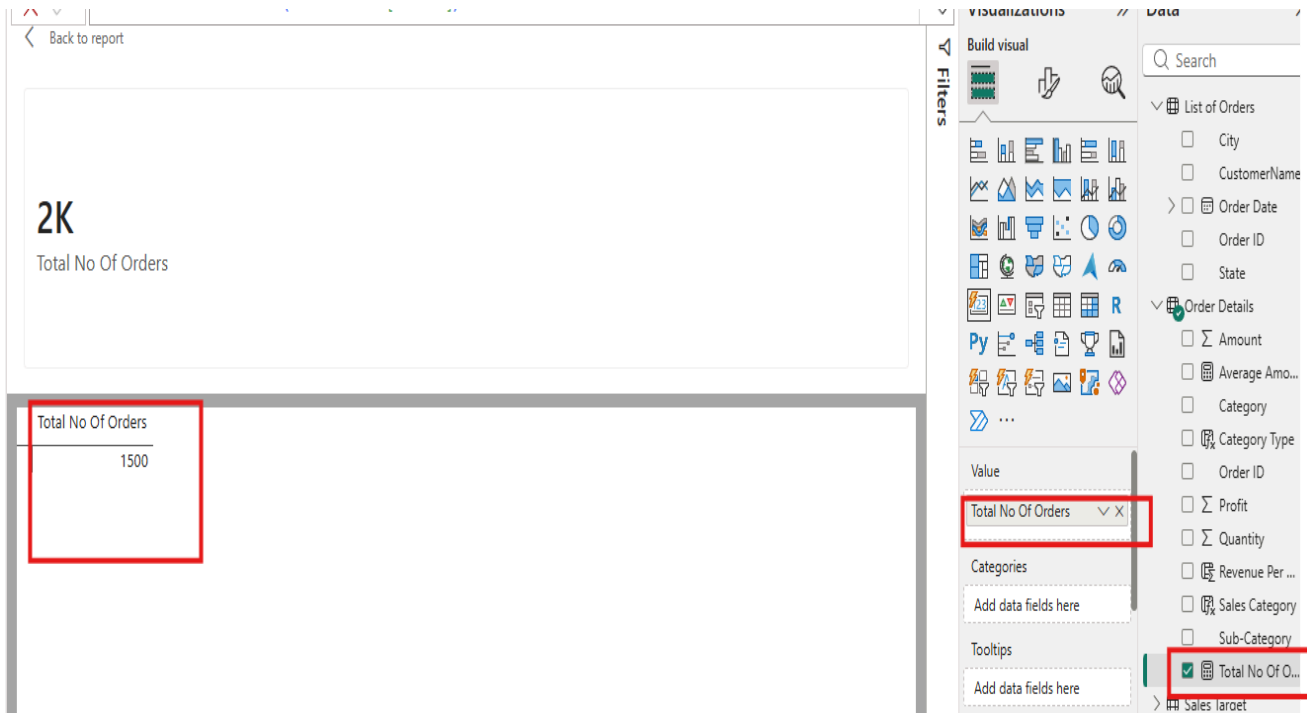
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The new calculated column is created using the Condition as in Order Details Table Column Named Amount is **Greater than Average** Amount 288 given the results as “**Above Average**”, Then rest will be resulted as “**Below Average**”. Then the Sales Category Column is created using the above conditions.

2.Calculated Measures:

1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.

The Calculated Measure is to Perform the given formula with the input we have given and for the output go to Visuals in that take card view will show the Results of the Performed Formula. There is no need of creating any column or Table only do calculations, and the stored under the current table we can get from the visibled Measure.



Order count is calculated using the Calculated Measure by Clicking the Table tools→select the New Measure and click on that we will get a formula bar in that type of the Formula as `Total No Of Orders = COUNTA('Order Details'[Order ID])`.

2. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.



Calculated Measure performed to calculate the average Profit gathered in Delhi to show how Profit is varied from other cities.

3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

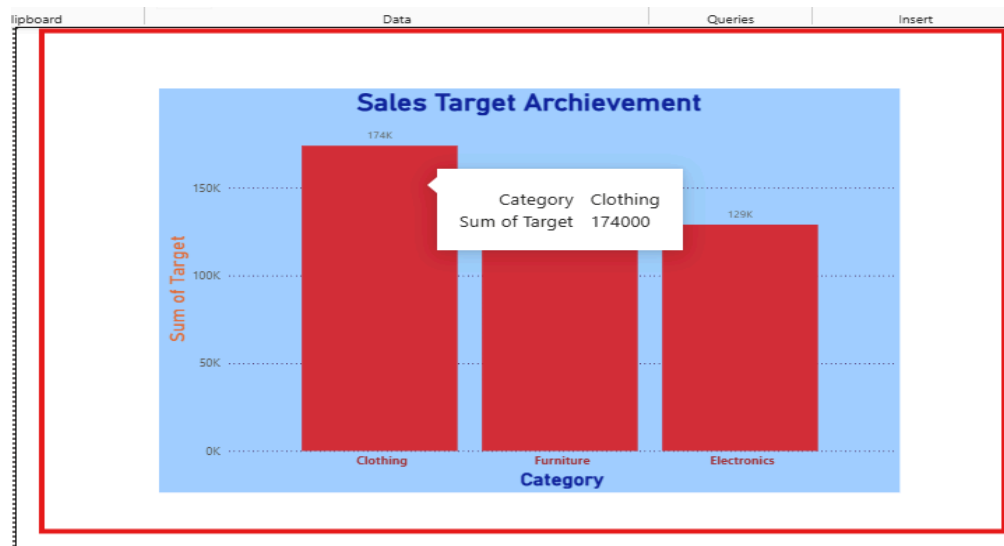


The Year to Date Calculated Measure to show the total sales accumulated from the starting date of the given data to current order date to know the sales updated data sales in the given dataset.

Data Visualization:

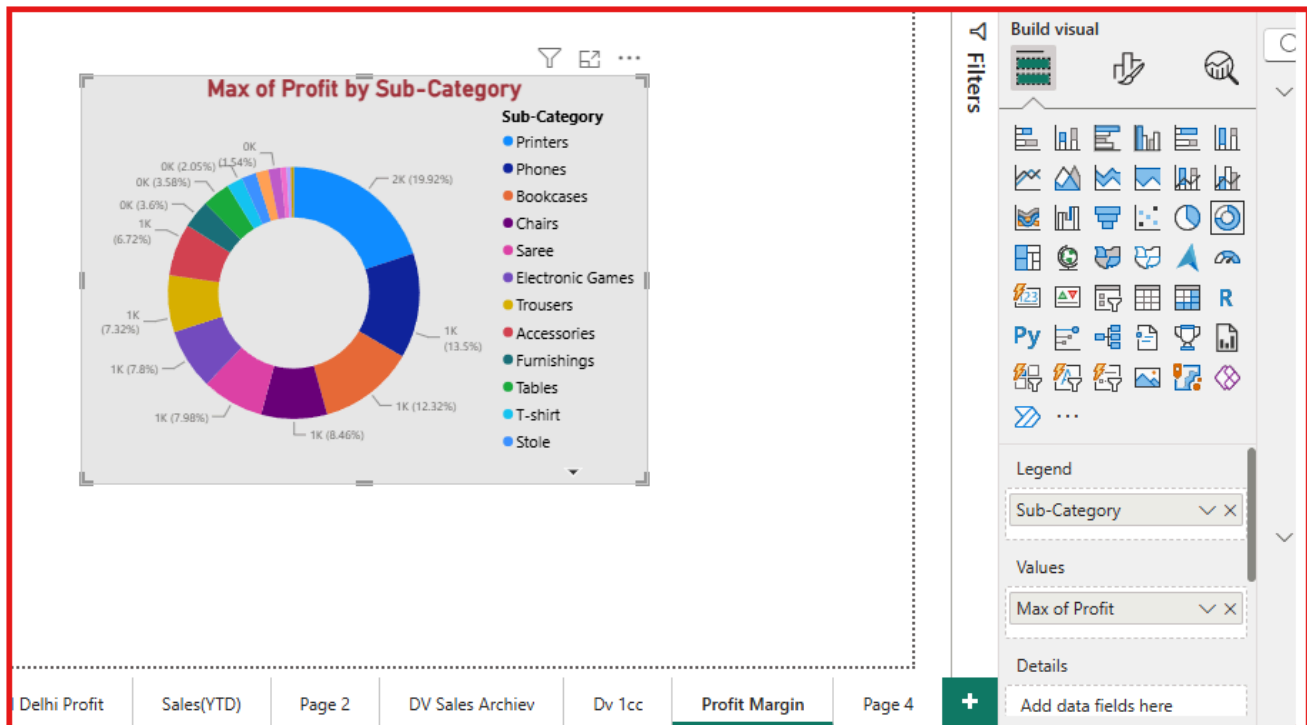
Create the following visualizations using Power BI: ● Sales Target

Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.



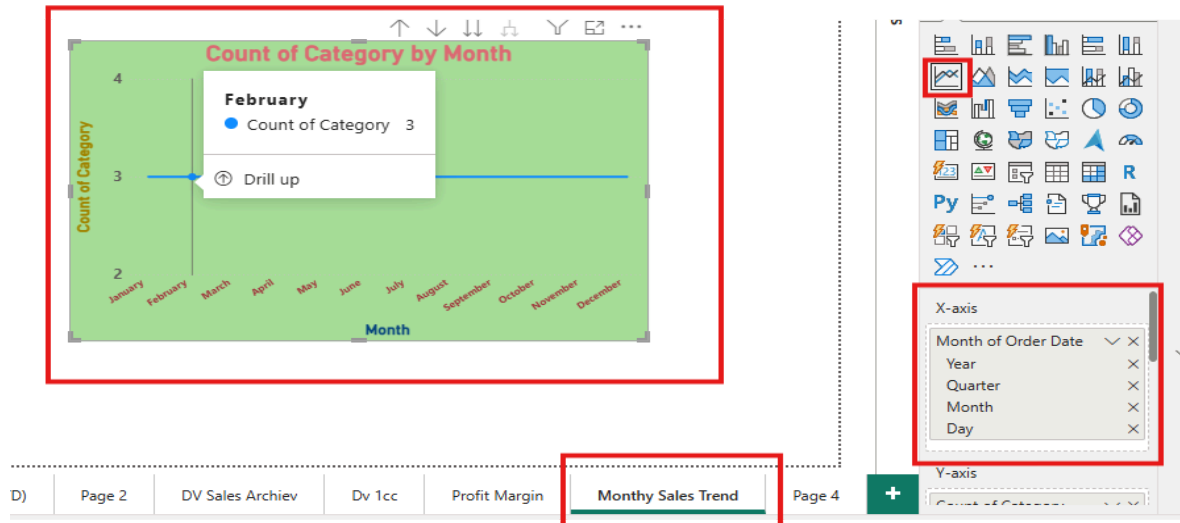
For Comparing the Sales with sales target and category we are taking the Clustered Column chart to Analyze the Sales target achievement. In this chart we are taking the x axis as a categorical column and y axis numerical value. In the above chart we get a Data Analysis of Clothing category given the high sales, so we can focus on the clothing to buy more to increase the sales Target.

- Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.



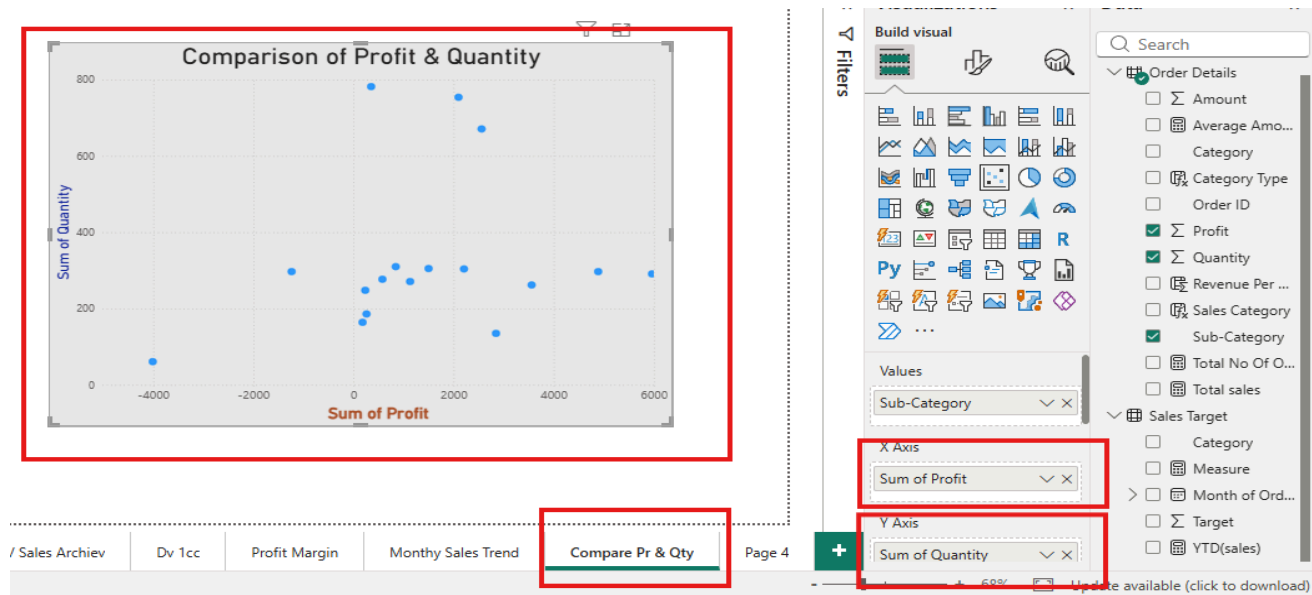
The Donut chart represents the Maximum profit of Sub-category by having sub category as Legend, Profit as Values field.

- Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



Line chart is for to show the time line also for the future prediction, here we can only pass the one numerical values as Y axis, one categorical data as X axis, if we want can get the secondary y axis.

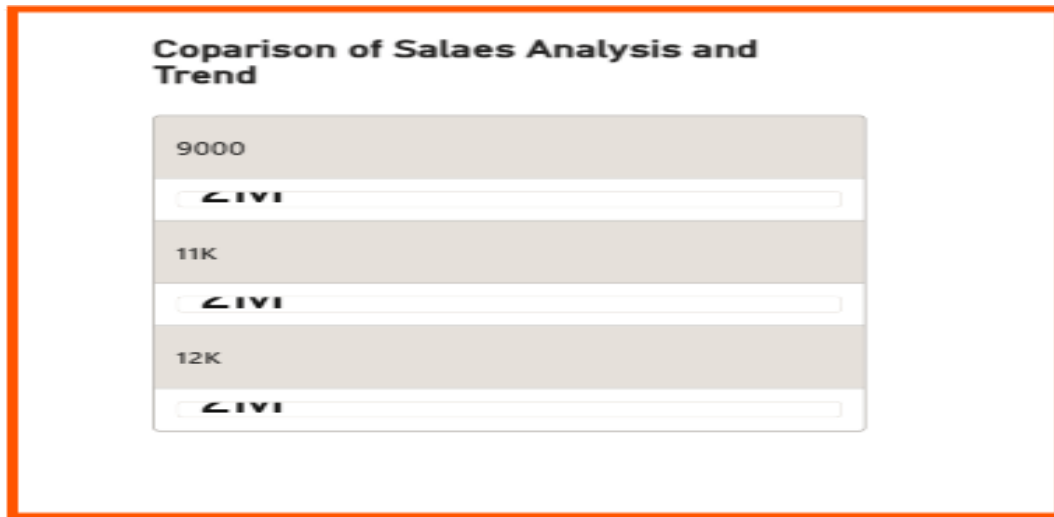
- Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



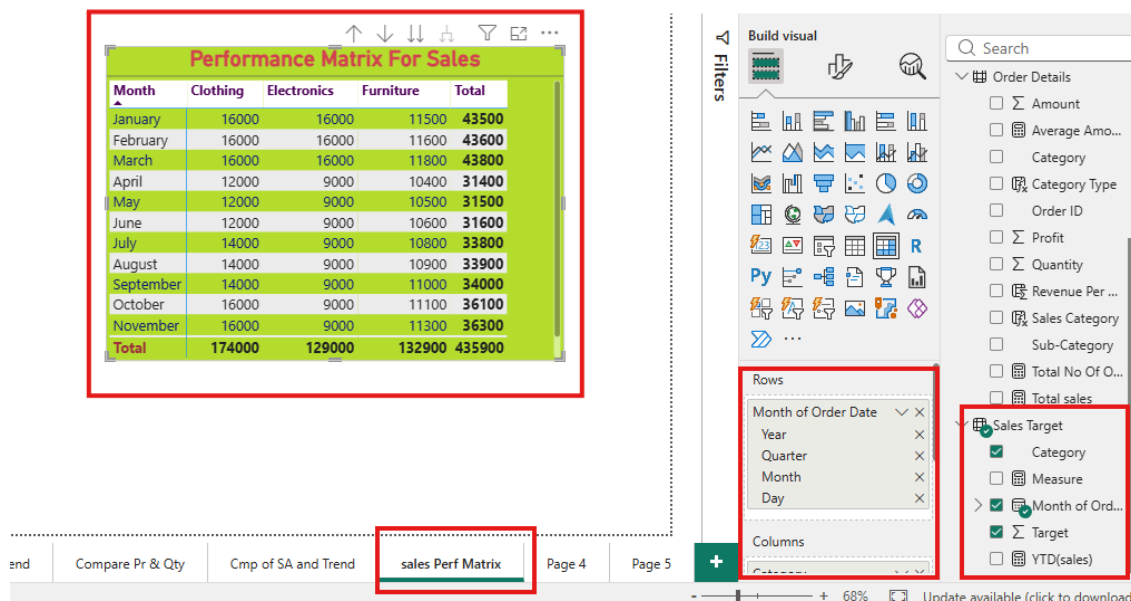
Scatterplot is mainly used for the visualization of two data variables in the Power BI. Here giving the values x axis as Profit and for Y axis Quantity by Sub-category to

compare and analyze the high profit where we are getting, here the High profit held in the level of 280-2000 is maximum high profit in subcategory in the sale data.

- Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

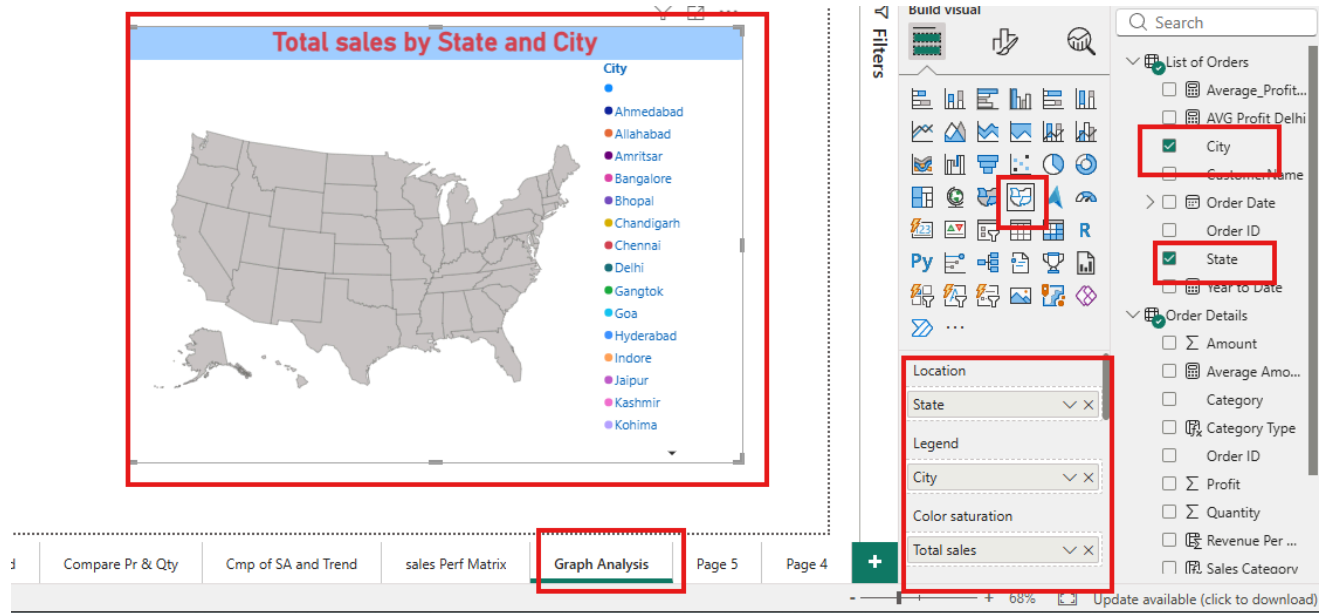


- Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



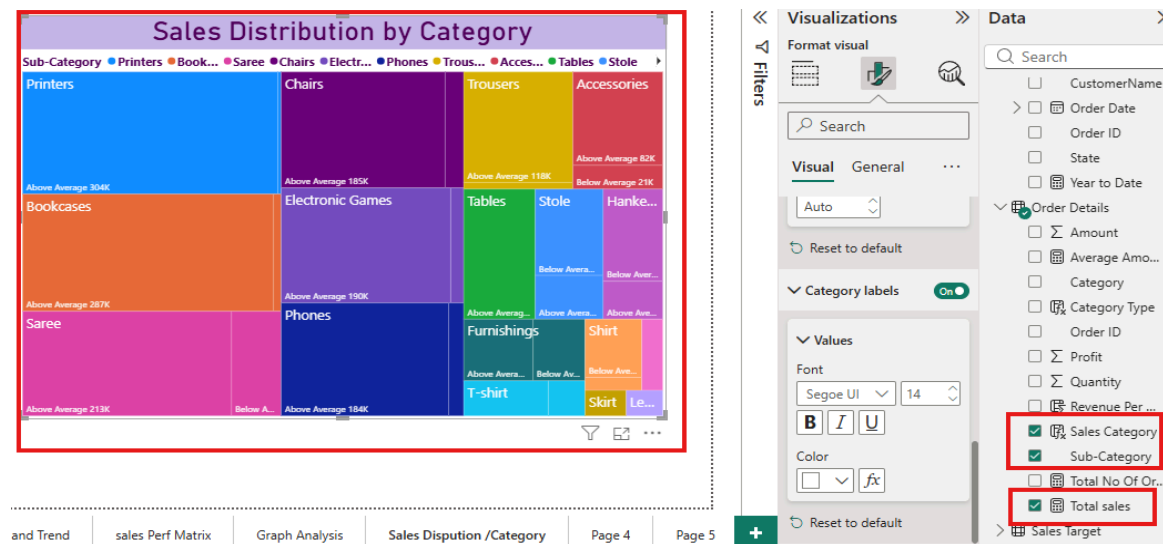
This matrix view chart is to show the sales data into multiple dimensions, and also it is similar to the pivot table and also having drill downs for the data table.

- **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.



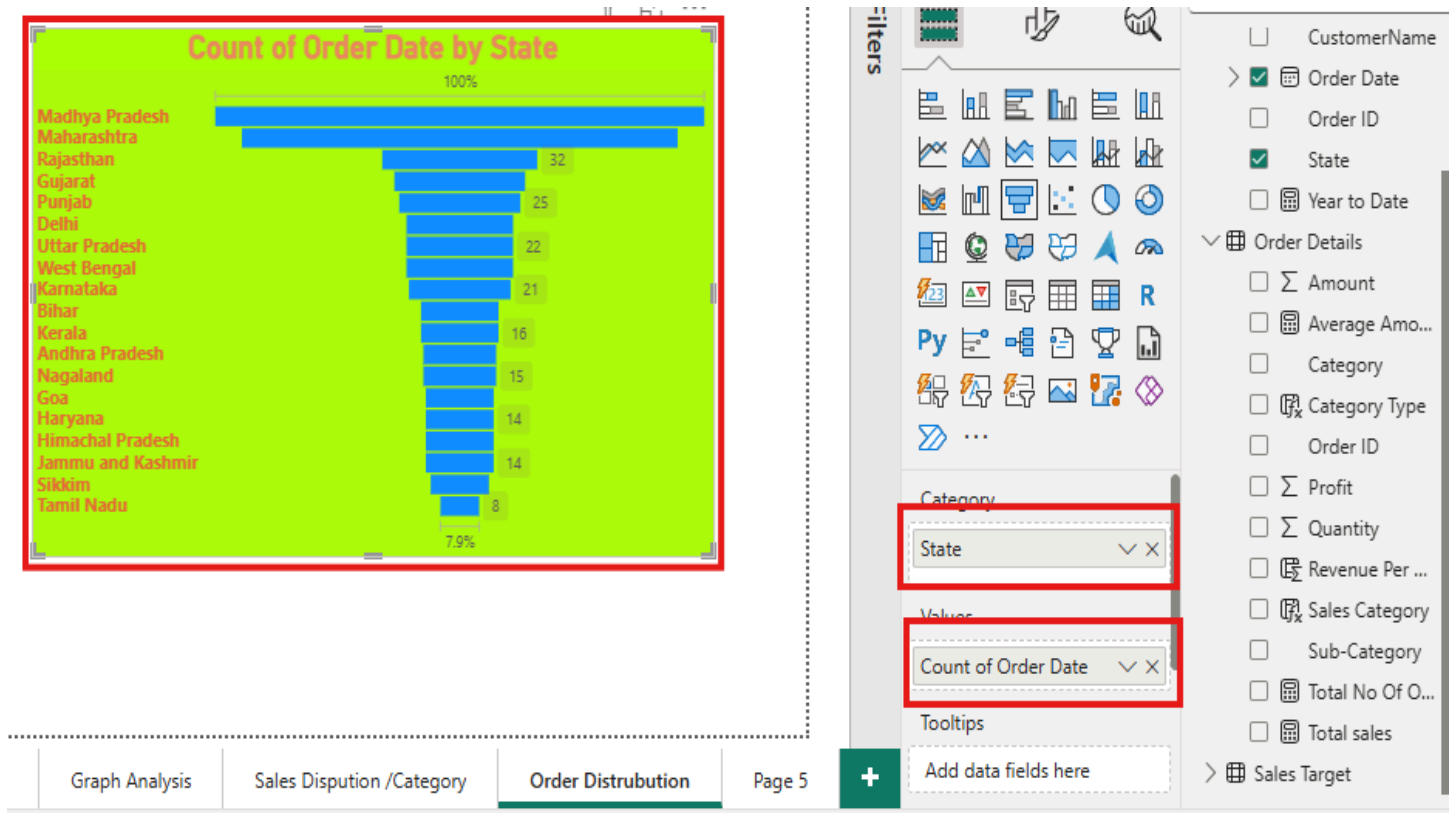
In the Geographic sales Analysis Shape map chart is to represent the data points of the cities Location. We can also mark it as a Bubble showed in the particular location.

- **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a treemap.



Tree map chart is a data visualization having multiple rectangles from Bigger to small as above image.

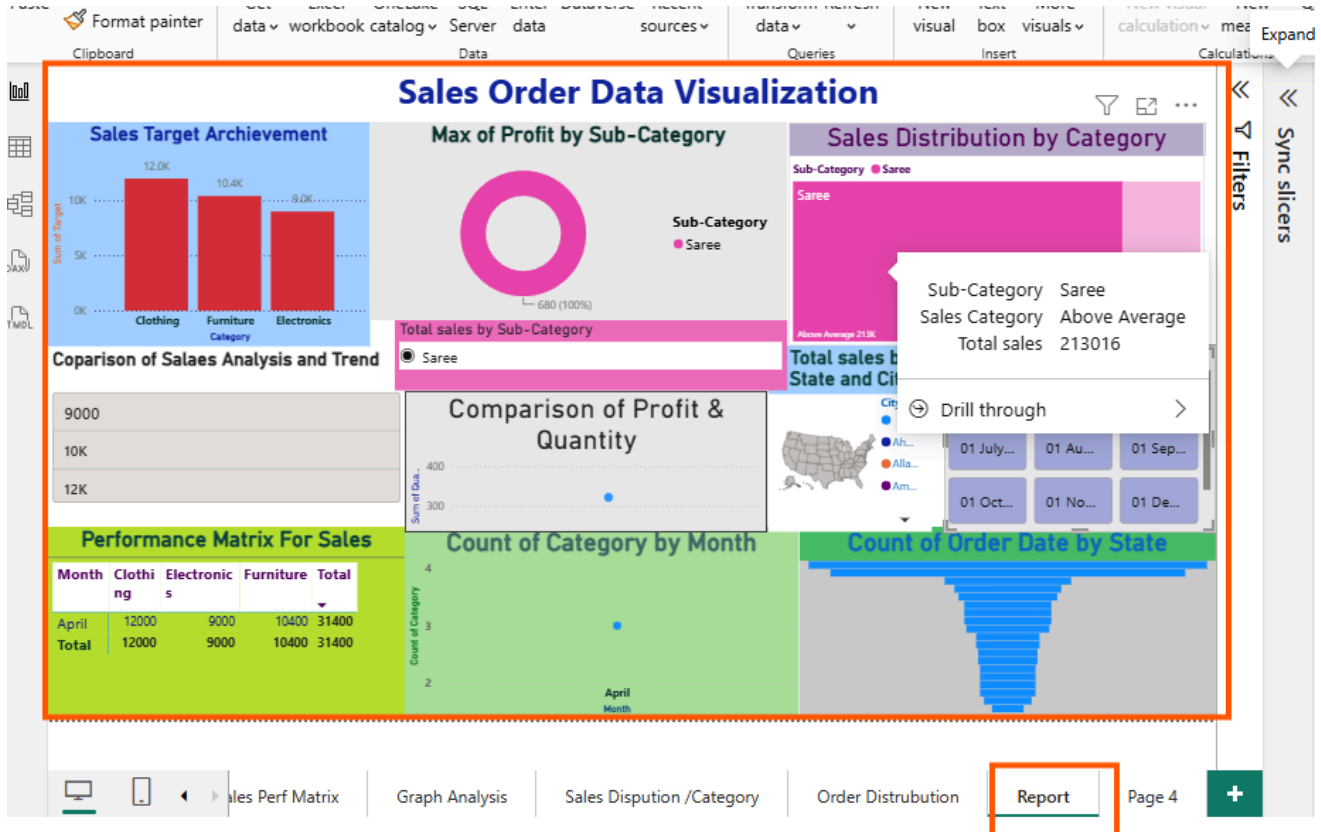
- **Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.**



The Funnel is to Visualize the Process of connected stages for conventional tracking. Here the tracking for the distribution of order is done by state wise of the ordered list.

States are listed as order count distribution among the states and high distribution in Madhya pradesh, likewise it is listed for visual.

Report of the Dax and Data Visualization:



The above visual represents the Sales Order ,Target Achievement,Profit of the whole orders data. This report shows the whole sales target which is achieved year wise Revenue per order. Sales performance metrics indicates category wise sales by knowing the sales date, month and Year.